



CAREERSENSE ACADEMY

Software QA & Product Testing

Functional testing, regression, browser compatibility, accessibility, performance, bug reporting, and release readiness

PARTNER LEARNING SERIES

Build the testing mindset required to find meaningful defects, reproduce them clearly, protect users, and help teams release software with confidence.

CareerSense Partners • developers • product teams • QA beginners • analysts

CareerSense Learning Series • careersenseai.com/resources

FRONT MATTER

How to Use This Handbook

Build the testing mindset required to find meaningful defects, reproduce them clearly, protect users, and help teams release software with confidence.

What each chapter contains

Each chapter follows the same learning rhythm used across the CareerSense Academy series: a concise introduction, learning objectives, plain-English concepts, CareerSense examples, professional habits, common mistakes, quick revision notes, practice questions, and hands-on exercises.

PARTNER PROMISE The examples are designed to prepare you for real CareerSense Partner assignments. Do not memorize terminology. Learn to create useful work products: audits, reports, prototypes, dashboards, campaigns, strategies, and decisions.

Recommended practice approach

- Read the relevant chapter before starting the matching Partner assignment.
- Use CareerSense or a safe sample environment to practise whenever possible.
- Keep evidence: screenshots, notes, links, spreadsheets, queries, designs, or metrics.
- Document assumptions and limitations instead of hiding uncertainty.
- Finish each chapter by completing at least one hands-on exercise.

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07	Performance and Reliability Testing
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Closing the book: quick revision sheet • glossary • professional habits • 20-question quiz • answer key • next steps.

CHAPTER 01

01

Quality as a Product Responsibility

Quality includes correctness, usability, reliability, performance, security, accessibility, compatibility, and maintainability.

LEARNING OBJECTIVES

- Define software quality
- Distinguish verification and validation
- Recognize quality dimensions
- Build a tester mindset

Core idea

Quality includes correctness, usability, reliability, performance, security, accessibility, compatibility, and maintainability.

Professional perspective

Testing is not only about proving a feature works. It is about discovering how it fails under realistic conditions.

CAREERSENSE CONNECTION

CareerSense QA should test whether users can complete important career workflows reliably across devices and browsers.

Practice lens	What it means
Weak approach	Start with activity or opinion without defining the desired outcome.
Professional approach	Define the decision, evidence, owner, success measure, and next action.
Evidence to keep	Notes, screenshots, metrics, links, artifacts, or stakeholder feedback.

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Define software quality.
- Distinguish verification and validation.
- Recognize quality dimensions.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

Q1 What is the main purpose of quality as a product responsibility?

Q2 How would you apply this topic to a CareerSense Partner assignment?

Q3 What evidence would you collect before making a recommendation?

Q4 What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to quality as a product responsibility using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 02

02

Test Planning and Test Cases

A test case should include preconditions, steps, test data, expected result, actual result, and status.

LEARNING OBJECTIVES

- Write test scenarios
- Create test cases
- Use positive and negative testing
- Prioritize risk

Core idea

A test case should include preconditions, steps, test data, expected result, actual result, and status.

Professional perspective

Risk-based testing focuses effort on high-impact and high-likelihood failures.

CAREERSENSE CONNECTION

Payment, downloadable reports, identity cards, certificates, and authentication deserve deeper testing than decorative UI elements.

Practice lens	What it means
Question to ask	What user or business problem are we trying to solve?
Risk to test	What assumption could make this work fail or produce little value?
Completion signal	What observable result proves the assignment is done well?

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Write test scenarios.
- Create test cases.
- Use positive and negative testing.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

- Q1** What is the main purpose of test planning and test cases?
- Q2** How would you apply this topic to a CareerSense Partner assignment?
- Q3** What evidence would you collect before making a recommendation?
- Q4** What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to test planning and test cases using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 03

03

Functional and Regression Testing

Functional testing validates expected behavior. Regression testing checks that new changes did not break previously working capabilities.

LEARNING OBJECTIVES

- Test business rules
- Understand regression suites
- Use boundary values
- Test state transitions

Core idea

Functional testing validates expected behavior. Regression testing checks that new changes did not break previously working capabilities.

Professional perspective

Boundary values often reveal defects: empty input, maximum length, zero credits, expired plans, or invalid dates.

CAREERSENSE CONNECTION

Test CareerSense plan transitions, credit depletion, top-ups, one-time free usage, and download payment behavior.

Practice lens	What it means
Weak approach	Start with activity or opinion without defining the desired outcome.
Professional approach	Define the decision, evidence, owner, success measure, and next action.
Evidence to keep	Notes, screenshots, metrics, links, artifacts, or stakeholder feedback.

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Test business rules.
- Understand regression suites.
- Use boundary values.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

- Q1** What is the main purpose of functional and regression testing?
- Q2** How would you apply this topic to a CareerSense Partner assignment?
- Q3** What evidence would you collect before making a recommendation?
- Q4** What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to functional and regression testing using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 04

04

Browser, Device, and Responsive Testing

Browser testing should define supported versions and critical flows. Responsive testing checks layout and interaction, not merely whether content fits.

LEARNING OBJECTIVES

- Build a compatibility matrix
- Test viewport behavior
- Recognize browser differences
- Capture reproducible evidence

Core idea

Browser testing should define supported versions and critical flows. Responsive testing checks layout and interaction, not merely whether content fits.

Professional perspective

Input behavior, file upload, PDF generation, fonts, and date handling often vary between environments.

CAREERSENSE CONNECTION

Run CareerSense core flows on Chrome, Safari, Edge, Firefox, and representative mobile sizes.

Practice lens	What it means
Question to ask	What user or business problem are we trying to solve?
Risk to test	What assumption could make this work fail or produce little value?
Completion signal	What observable result proves the assignment is done well?

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Build a compatibility matrix.
- Test viewport behavior.
- Recognize browser differences.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

Q1 What is the main purpose of browser, device, and responsive testing?

Q2 How would you apply this topic to a CareerSense Partner assignment?

Q3 What evidence would you collect before making a recommendation?

Q4 What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to browser, device, and responsive testing using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 05

05

Bug Reporting and Triage

A useful bug report includes title, environment, preconditions, exact steps, expected result, actual result, reproducibility, evidence, and impact.

LEARNING OBJECTIVES

- Write reproducible defects
- Use severity and priority
- Capture logs and screenshots
- Communicate without blame

Core idea

A useful bug report includes title, environment, preconditions, exact steps, expected result, actual result, reproducibility, evidence, and impact.

Professional perspective

Severity describes technical/user impact; priority describes urgency for the business.

CAREERSENSE CONNECTION

A broken certificate download affecting every paid student may be both high severity and high priority.

Practice lens	What it means
Weak approach	Start with activity or opinion without defining the desired outcome.
Professional approach	Define the decision, evidence, owner, success measure, and next action.
Evidence to keep	Notes, screenshots, metrics, links, artifacts, or stakeholder feedback.

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Write reproducible defects.
- Use severity and priority.
- Capture logs and screenshots.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

Q1 What is the main purpose of bug reporting and triage?

Q2 How would you apply this topic to a CareerSense Partner assignment?

Q3 What evidence would you collect before making a recommendation?

Q4 What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to bug reporting and triage using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 06

06

Accessibility and Usability Testing

Accessibility is part of quality. Test whether users can navigate with keyboard, understand form labels, perceive focus, and recover from errors.

LEARNING OBJECTIVES

- Check keyboard navigation
- Understand contrast and labels
- Test focus and error messages
- Identify usability friction

Core idea

Accessibility is part of quality. Test whether users can navigate with keyboard, understand form labels, perceive focus, and recover from errors.

Professional perspective

Usability testing observes whether a user can complete a task without confusion.

CAREERSENSE CONNECTION

CareerSense forms should provide clear field labels, meaningful errors, visible focus, and logical tab order.

Practice lens	What it means
Question to ask	What user or business problem are we trying to solve?
Risk to test	What assumption could make this work fail or produce little value?
Completion signal	What observable result proves the assignment is done well?

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Check keyboard navigation.
- Understand contrast and labels.
- Test focus and error messages.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

- Q1** What is the main purpose of accessibility and usability testing?
- Q2** How would you apply this topic to a CareerSense Partner assignment?
- Q3** What evidence would you collect before making a recommendation?
- Q4** What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to accessibility and usability testing using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 07

07

Performance and Reliability Testing

Performance testing includes load time, response time, rendering, network behavior, and resource usage.

LEARNING OBJECTIVES

- Read Lighthouse outputs
- Understand latency and throughput
- Test failures and retries
- Recognize reliability scenarios

Core idea

Performance testing includes load time, response time, rendering, network behavior, and resource usage.

Professional perspective

Reliability testing intentionally explores failures such as slow APIs, timeouts, duplicate clicks, and interrupted uploads.

CAREERSENSE CONNECTION

Measure CareerSense landing and dashboard pages, then test what happens when an AI request is slow or fails.

Practice lens	What it means
Weak approach	Start with activity or opinion without defining the desired outcome.
Professional approach	Define the decision, evidence, owner, success measure, and next action.
Evidence to keep	Notes, screenshots, metrics, links, artifacts, or stakeholder feedback.

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Read Lighthouse outputs.
- Understand latency and throughput.
- Test failures and retries.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

- Q1** What is the main purpose of performance and reliability testing?
- Q2** How would you apply this topic to a CareerSense Partner assignment?
- Q3** What evidence would you collect before making a recommendation?
- Q4** What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to performance and reliability testing using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CHAPTER 08

08

CareerSense Capstone: 360° Quality Audit

Audit every major CareerSense service using a traceable test matrix. Include functional, compatibility, responsive, accessibility, performance, and usability observations.

LEARNING OBJECTIVES

- Create a QA plan
- Execute cross-browser testing
- Report defects
- Produce a release recommendation

Core idea

Audit every major CareerSense service using a traceable test matrix. Include functional, compatibility, responsive, accessibility, performance, and usability observations.

CAREERSENSE CONNECTION

Deliver test cases, defect log, evidence, severity summary, top 10 improvements, and a go/no-go recommendation.

Practice lens	What it means
Question to ask	What user or business problem are we trying to solve?
Risk to test	What assumption could make this work fail or produce little value?
Completion signal	What observable result proves the assignment is done well?

COMMON MISTAKE

Producing a polished artifact without showing the evidence, assumptions, decision logic, and measurable outcome behind it.

QUICK REVISION NOTES

- Create a QA plan.
- Execute cross-browser testing.
- Report defects.
- Tie recommendations to evidence and expected impact.
- Communicate constraints and uncertainty clearly.

Practice Questions

Q1 What is the main purpose of careersense capstone: 360° quality audit?

Q2 How would you apply this topic to a CareerSense Partner assignment?

Q3 What evidence would you collect before making a recommendation?

Q4 What is one common mistake a beginner should avoid?

Hands-on Exercises

- 1 Create a one-page working artifact related to careersense capstone: 360° quality audit using CareerSense or a safe sample scenario.
- 2 Review your artifact as if you were a startup leader: identify one missing assumption, one risk, and one measurable improvement.
- 3 Write a 150-word executive summary explaining what you found and what should happen next.

CLOSING THE BOOK

Course Summary

You have worked through the professional foundations of software qa & product testing. The purpose of this handbook is not to make you fluent in vocabulary; it is to help you complete real work with stronger judgment, structure, evidence, and communication.

**PROFESSIONAL
TAKEAWAY**

CareerSense Partners should be able to explain not only what they did, but why it mattered, what evidence supported it, what trade-offs existed, and what measurable action should happen next.

Software QA & Product Testing - Quick Revision Sheet

Topic	Remember
Quality as a Product Responsibility	Define software quality
Test Planning and Test Cases	Write test scenarios
Functional and Regression Testing	Test business rules
Browser, Device, and Responsive Testing	Build a compatibility matrix
Bug Reporting and Triage	Write reproducible defects
Accessibility and Usability Testing	Check keyboard navigation
Performance and Reliability Testing	Read Lighthouse outputs
CareerSense Capstone: 360° Quality Audit	Create a QA plan

Glossary

Term	Plain-English meaning
PRD	Product Requirements Document describing problem, scope, behavior, and success.
Acceptance criteria	Testable conditions that define expected behavior.
Roadmap	A prioritized view of intended outcomes and initiatives over time.
Regression	Retesting existing behavior after changes.
Severity	Impact of a defect on users or systems.
Test case	Defined preconditions, steps, data, and expected result.
Outcome	A measurable change in user, customer, or business behavior.
Metric	A quantitative measure used to monitor performance or progress.
Hypothesis	A testable statement about what change is expected and why.
Stakeholder	A person or group affected by or able to influence an initiative.
MVP	The smallest trustworthy version that tests an important assumption.
KPI	A key performance indicator tied to an important objective.
Iteration	A cycle of building, measuring, learning, and improving.
Trade-off	Choosing one benefit or priority while accepting a cost elsewhere.
Evidence	Observed information that supports or challenges a decision.
Prioritization	Deciding what deserves attention and resources first.

20-Question Practice Quiz

1. What is one core principle from 'Quality as a Product Responsibility'?
9. How could a CareerSense Partner apply 'Quality as a Product Responsibility' in a real assignment?
2. What is one core principle from 'Test Planning and Test Cases'?
10. How could a CareerSense Partner apply 'Test Planning and Test Cases' in a real assignment?
3. What is one core principle from 'Functional and Regression Testing'?
11. How could a CareerSense Partner apply 'Functional and Regression Testing' in a real assignment?
4. What is one core principle from 'Browser, Device, and Responsive Testing'?
12. How could a CareerSense Partner apply 'Browser, Device, and Responsive Testing' in a real assignment?
5. What is one core principle from 'Bug Reporting and Triage'?
13. How could a CareerSense Partner apply 'Bug Reporting and Triage' in a real assignment?
6. What is one core principle from 'Accessibility and Usability Testing'?
14. How could a CareerSense Partner apply 'Accessibility and Usability Testing' in a real assignment?
7. What is one core principle from 'Performance and Reliability Testing'?
15. How could a CareerSense Partner apply 'Performance and Reliability Testing' in a real assignment?
8. What is one core principle from 'CareerSense Capstone: 360° Quality Audit'?
16. How could a CareerSense Partner apply 'CareerSense Capstone: 360° Quality Audit' in a real assignment?
17. Why should recommendations be tied to evidence?
18. What is the difference between an activity and an outcome?
19. Why should assumptions and limitations be documented?
20. What makes an assignment deliverable useful to leadership?

Answer Key

1. A strong answer should explain define software quality and connect it to an observable decision or result.
9. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
2. A strong answer should explain write test scenarios and connect it to an observable decision or result.
10. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
3. A strong answer should explain test business rules and connect it to an observable decision or result.
11. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
4. A strong answer should explain build a compatibility matrix and connect it to an observable decision or result.
12. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
5. A strong answer should explain write reproducible defects and connect it to an observable decision or result.
13. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
6. A strong answer should explain check keyboard navigation and connect it to an observable decision or result.
14. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
7. A strong answer should explain read lighthouse outputs and connect it to an observable decision or result.
15. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
8. A strong answer should explain create a qa plan and connect it to an observable decision or result.
16. Apply the concept using a specific CareerSense workflow, evidence, deliverable, and measurable outcome.
17. Evidence separates a useful recommendation from an unsupported opinion.
18. An activity is work performed; an outcome is the change produced by that work.
19. Documenting assumptions makes uncertainty visible and allows reviewers to judge the strength of the conclusion.
20. A useful deliverable is clear, evidence-based, actionable, appropriately scoped, and tied to an owner or next decision.

CAREERSENSE ACADEMY

Keep Building

You now have a working foundation in software qa & product testing. Use this handbook as a reference before and during your Partner assignments. Professional capability grows when you repeatedly turn concepts into useful outputs, receive feedback, and improve your judgment.

NEXT STEPS

- Choose the CareerSense Partner assignment most closely related to this handbook.
- Review the relevant chapter before beginning.
- Complete the assignment with traceable evidence.
- Ask for feedback on both the result and your decision process.
- Add the finished work to your Partner portfolio where permitted.

**CAREERSENSE
PARTNER
LEARNING PATH**

Learn → Apply → Document → Review → Improve → Lead.

More CareerSense learning resources

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